

1 Changes from the Sketch to the Formal Ontology — Art Haxholli

Ferroelectric HfO₂ Memory

This document records what changed between the ontology specification as sketched and the formal OWL/Turtle model, and the reason for each change. The changes fall into four groups: corrections found by reading the specification against its own instance document, one structural change to how layer stacks are represented, changes prompted by reviewing the models produced by the other workstream members, and decisions that only became necessary once the specification was executed as OWL.

The resulting file contains 3,933 triples and is consistent under HermiT.

1.1 Corrections Internal to the Specification

These were found by checking the specification against the instance document, which describes the same flows at instance level. They are errors in the earlier version rather than consequences of anything external.

The data property `hasValue` on `Measurement` was renamed to `hasMeasuredValue`. The original name collides with `owl:hasValue`, a reserved OWL construct. A property with that local name in a private namespace is a distinct IRI and therefore legal, but the collision is confusing to read and unnecessary.

The property `hasCycleCount` was declared twice, on `Measurement` with range `xsd:long` and on `Unit_Process` with range `xsd:int`. In OWL these are one property with two range axioms, and the two meanings are unrelated: one counts endurance cycles during characterisation, while the other counts ALD supercycles during deposition. The property was split into `hasEnduranceCycleCount` and `hasDepositionCycleCount`.

The specification declared `hasTrapDensity_cm2` as a data property on `Material`, carrying S10's values of 1×10^{14} and $3.6 \times 10^{14} \text{ cm}^{-2}$. The instance document modelled the same two figures as separate `Measurement` instances on one specimen at two points in its cycling history. The two documents therefore contradicted each other.

The instance document was correct: a data property on the material can hold only one of the two values and loses the cycling history entirely. The conflict was resolved by stating a general rule: measured quantities become `Measurement` instances, while specified or nominal quantities remain data properties.

Seven properties moved off `Material` as a result: trap density, film thickness, carbon impurity, film density, grain length and width, the measured branch of the Hf:Zr ratio, and crystallisation onset.

The classes `Artifact` and `Technology_Node` were used as ranges of `hasInput`, `hasOutput`, `hasMeasurement`, `measuredOn`, and `flowTargetsNode`, but did not appear in the class tree. Both were added. `Artifact` was given four subclasses corresponding to the specimen types already named in the specification: blanket film, patterned capacitor, single cell, and array.

The properties `hasTopElectrode`, `hasBottomElectrode`, and `layerIndex` were referenced only in the Notes columns of property tables and were never declared. Each was given its own declaration.

`hasStepIndex` appeared among the object properties with neither domain nor range. It was moved to the data properties as an `xsd:int` on `Unit_Process`, with a note that it is a convenience key rather than the authoritative ordering, which is carried by `belongsToFlow` together with `directlyPrecedes`.

The verified ordering facts were written at class level, in the form

`directlyPrecedes(Top_Electrode_Deposition, Crystallisation_Anneal).`

These are generalisations over per-flow individuals, not axioms between classes. Asserting `precedes` between classes would require punning and would be incoherent given that Flow C orders the same two class-level steps in the opposite direction. The statements were therefore restated as holding between the step individuals of each named flow.

Finally, the specification recorded in prose that line-of-sight deposition blocks the 3D capacitor, but stated this only as a note on the `enables` property, leaving it unassertable. A `precludes` property was added.

1.2 Structural Change: Reification of the Layer Slot

The specification left an open choice between `hasLayerAtPosition` and an RDF list for representing stack order. The choice was closed by introducing a `Layer_Slot` class.

One slot represents one position in one stack and carries:

- the `layerIndex`,
- the material occupying the position,
- the role that material serves there, and
- the as-fabricated thickness.

A `Layer_Role` class holds the eleven roles as a closed enumeration.

Three separate problems in the sources motivated this design, and none of them is solved by a direct link from stack to material.

First, role is positional rather than substantial. `Ti_Pt_contact` is an `Electrode_Material` by class but functions as a probe pad in the Flow A coupon, while platinum is a functional electrode in the PLD and CSD stacks.

Second, one material can occupy different roles in different flows. Silicon is a substrate in Flow A and the channel in Flow B1.

Third, thickness varies per stack while the material remains the same. HZO is 9 nm in Flow B1 and 9.5 nm in Flow G, and neither figure is a property of the substance itself.

The role-typed properties `hasElectrode`, `hasTopElectrode`, `hasBottomElectrode`, `hasInterfacialLayer`, and `hasFerroelectricLayer` were retained but redefined as derived shortcuts computed from the slot roles, with the slot treated as authoritative.

For the same reason, the loose data properties `hasPhaseFraction`, `fractionMethod`, and `phasesLumped` were removed from `Phase_State` and folded into `Measurement`.

A phase fraction is produced by a named method on a named specimen and is not interpretable without its lumping convention. This is the same value-plus-conditions structure that the `Measurement` class already exists to represent.

1.3 Changes Prompted by the Other Workstream Models

Four changes came from reviewing the SMT2020 and ferroelectric-memory models produced by the other two workstream members.

1.3.1 Splitting Degradation_Phenomenon

`Degradation_Phenomenon` was split into `Film_Level_Degradation` and `Array_Level_Degradation`.

The SMT2020 model by Fatos introduces a `Downtime` superclass over `PM` and `Breakdown`, where the previous sketch had two independent sibling classes. That structure made it apparent that a flat sibling set can conceal a distinction worth stating.

In this ontology the equivalent case is `Degradation_Phenomenon`, whose children were flat even though two of them differ fundamentally from the rest. `Pass disturb` and `read disturb` only exist once cells are wired

into an array, while wake-up, fatigue, imprint, and retention loss are properties of the film. Making this distinction explicit also exposed a defect in the existing model. `exhibits` had domain

`Memory_Cell_Architecture` $\cup$ `Layer_Stack`,

which provided no possible subject for `Pass_Disturb`. The domain was widened to include `Array_Architecture`.

1.3.2 Adding Characterisation

`Characterisation` was added as a subclass of `Unit_Process`.

The ferroelectric-memory model by Olti places a `Quality_Check` node inside the process chain, connected to the output through `validates` and `verifiedBy`. In this specification, `Measurement_Method` and `Measurement` sat beside the process flow rather than within it.

That arrangement left `Metrology_Tool` unreachable. `usesEquipment` has domain `Unit_Process`, and no subclass of `Unit_Process` performed measurement, so no tool of that kind could be attached to anything.

`Characterisation` was therefore added with two subclasses, structural and electrical, together with a `producesMeasurement` property linking the step to its results.

A characterisation step now carries `belongsToFlow`, occupies a position in the ordering, and reaches its instruments through the ordinary equipment property.

1.3.3 Scope of FerroelectricMaterial

A scope note was added to `FerroelectricMaterial`.

Olti's model covers ferroelectrics broadly, including perovskites, multiferroics, and hydrogen-bond ferroelectrics. This ontology is restricted to the HfO_2 family by design, but the class name did not state this and every subclass was HfO_2 -based, leaving the restriction implicit.

A `scopeNote` annotation now records what is excluded and makes clear that the exclusion is deliberate rather than a gap in evidence.

The class retains its general name so that its scope can be widened later without requiring a rename.

1.3.4 Documenting the Modelling Conventions

Both other models state their class definitions inside the ontology itself, whether through `rdfs:comment` or `dcterms:description`. This specification kept its conventions in prose that the OWL file would not otherwise carry.

A conventions section was therefore added covering:

- the naming rule,
- the annotation properties,
- the measured-versus-specified rule,
- the disjointness policy, including pairs deliberately left non-disjoint, and
- the open-world reading.

The five annotation properties

`sourceLocator`, `epistemicStatus`, `uncertaintyRef`,
`contradictedBy`, and `scopeNote`

were declared first in the build so that every subsequent entity could carry them.

1.4 Decisions Forced During Formalisation

Two conflicts surfaced only when the specification was executed as OWL.

The specification gave `precedes` the characteristics transitive, asymmetric, and irreflexive. That combination is not permitted in OWL 2 DL, where asymmetry and irreflexivity may only be declared on simple properties and a transitive property is, by definition, non-simple.

A reasoner does not report this as an inconsistency; it declines to load the ontology.

Transitivity was kept because the transitive closure is what derives ordering facts from a step chain. Asymmetry and irreflexivity were removed. The flows are generated as linear chains, so no cycle can arise from the data. If the constraint is required later, it belongs in a SHACL shape rather than in OWL.

`directlyPrecedes` is simple and retains both characteristics.

The class tree drew the crystalline phases, defect types, and measurement methods as subclasses, while the instance document typed them as individuals.

The causal properties settle the question. For example,

`stabilisesPhase`

relates `Oxygen_Vacancy` to `Orthorhombic_Pca21`, while

`measurementMethod`

relates a measurement to `IdVg_Transfer`. Object properties relate individuals, so these entries were formalised as individuals.

The subclasses of `Degradation_Phenomenon` were left as classes in order to preserve the film-level and array-level split described above. This means the `exhibits` assertions pun on them. The result remains inconsistent with the rule applied elsewhere and is recorded as an open item rather than resolved.

Five further decisions were required and are documented in full alongside the generator.

Steps whose technique the source does not name are typed at the parent class rather than guessed at, following the principle already stated in the instance document for `A3_FE_Deposition`.

`Si_substrate` is typed once as `Substrate`, with its two roles carried by slots. This is why `Substrate` and `Channel_Material` are left non-disjoint.

The derived electrode shortcuts are emitted by the generator from the same source data as the slots rather than inferred using SWRL rules.

`slotMaterial` and `slotRole` carry qualified exact-cardinality restrictions in addition to being functional.

Finally, a second artifact individual was introduced for the cycled FeFET specimen because the instance document attached both trap-density measurements to one specimen while `measuredOn` is functional.

1.5 Result

The generated ontology contains:

- 3,933 triples,
- 322 individuals,
- eight process flows,
- 62 process steps,
- 34 layer slots,
- four specimens, and
- six measurements.

Provenance coverage is complete. A total of 447 entities carry a **sourceLocator**, and every remaining entity carries an explicit **epistemicStatus** of inferred, scaffolding, or named-only.

The status distribution is:

- 437 verified,
- 8 uncertain,
- 17 inferred,
- 102 scaffolding, and
- 1 named-only.

Verification covers reasoner consistency, referential integrity of all domains and ranges, provenance coverage, flow structure including the requirement that **precedes** never crosses a flow boundary, the slot design, the measurement reification, and agreement with the instance document.

All checks pass.

Three items remain open and are recorded rather than resolved:

- the punning on the subclasses of **Degradation_Phenomenon**,
- the single member of **Technology_Node**, and
- the empty **Measurement**, **Artifact**, and **Metrology_Tool** sets for Flow A, which reflect the absence of any characterisation of a Flow A sample in the source rather than a gap in the transcription.